

Unit 2 2. Vectors

2.1 Vectors and Types of Vectors

After completing this section, you should be able to:

- describe the difference between vector and scalar quantities.
- list down the common vector quantities in our everyday life.
- discuss geometric representation of vectors.
- give the definitions of the different types of vectors.

2.2 Graphical Method of Addition of Vectors in Two Dimensions (2-D)

After completing this section, you should be able to:

- define the term resultant vector.
- explain the geometric method for addition and subtraction of vectors in a plane.
- apply geometric method of addition of vectors to find resultant of vectors in two dimensions.

2.3 Algebraic Method of addition of vectors in Two Dimensions (2-D)

After completing this section, you should be able to:

- apply algebraic method to find the resultant of two collinear vectors, and perpendicular vectors.
- apply trigonometry to determine the direction of a resultant vector in terms of
- the angle it forms with a reference direction.
- compare the resultant of two given vectors as determined by way of geometric
- construction and by analytic method.
- resolve a vector into its components.
- find the resultant of two or more vectors applying component method.
- define unit vector and determine a unit vector in the direction of a given vector.

- apply the unit representation of vectors to determine the resultant of two or more vectors.

2.4 Product of Vectors

After completing this section, you should be able to:

- define dot product of a vector
- apply the technique of dot product to solve practical problems